ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Sensors and closed-loop speed control; relay and PLC control of two conveyors; hydraulic pumps, valve symbols and a solenoid-valve circuit; electro-hydraulic jog control with interlocking; and fault-finding.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw all circuits and symbols neatly, using a ruler and standard symbols.
5. Use only a black or blue pen.
6. Write neatly and legibly.

QUESTION 1

1.1 Define the term *sensor*. (2)

1.2 Name THREE types of sensors used in mechatronic systems and state what EACH one detects. (3)

1.3 Differentiate between an *inductive* and a *capacitive* proximity sensor. (2 × 2) (4)

1.4 Draw a labelled block diagram of a closed-loop system that controls the speed of a DC motor. (5)

1.5 A temperature transmitter gives an output of 4 mA at 0 °C and 20 mA at 100 °C. Calculate the output current at 60 °C. (6)

[20]

QUESTION 2

2.1 Explain the function of a soft starter on an electric motor. (2)

2.2 State THREE advantages of a PLC over hard-wired relay control. (3)

2.3 Explain the function of an on-delay timer in a control circuit. (2)

2.4 FIGURE 1 below shows the relay control circuit of two conveyors.

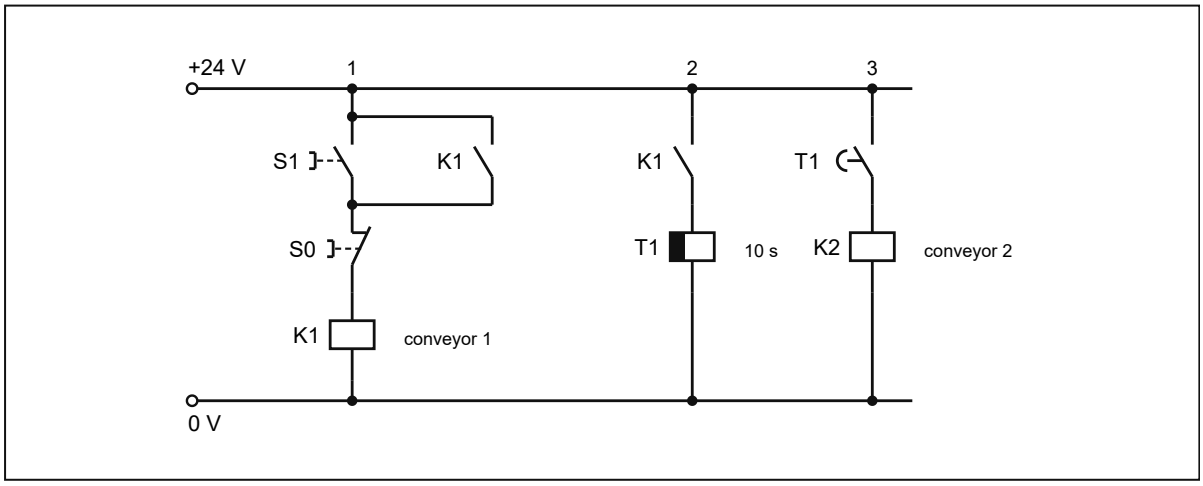


FIGURE 1

2.4.1 Name the component S0. (1)

2.4.2 Explain what happens when S1 is pressed and released. (2)

2.4.3 Explain the purpose of the K1 contact in current path 2. (2)

2.4.4 Explain when conveyor 2 starts. (2)

2.5 Draw the PLC ladder diagram that does the same as the circuit in FIGURE 1. Use I0.0 for S1, I0.1 for S0 (wired NC), Q0.0 for conveyor 1, Q0.1 for conveyor 2 and timer T1. (6)

[20]

QUESTION 3

3.1 Name THREE types of hydraulic pumps and give ONE advantage of a gear pump. (4)

3.2 Draw the symbol of EACH of the following:

3.2.1 4/3-way directional control valve, solenoid operated on both sides, spring centred, with a closed centre (4)

3.2.2 Variable-displacement hydraulic pump (one direction of flow) (2)

3.3 Design a hydraulic circuit for a double-acting cylinder controlled by the valve in QUESTION 3.2.1 (Y1 extends, Y2 retracts). The cylinder must extend slowly (meter-out control). Include a hydraulic power pack and a pressure gauge. (10)

[20]

QUESTION 4

4.1 Explain the function of a proximity sensor in an electro-hydraulic system. (2)

4.2 Name FOUR safety devices that are used in electro-hydraulic control circuits. (4)

4.3 A double-acting cylinder is controlled by a 4/3-way double-solenoid valve (Y1 extends, Y2 retracts). Design the electrical control circuit so that: the cylinder extends only while push button S1 is held in and retracts only while push button S2 is held in; Y1 and Y2 can never be energised at the same time (use the NC contacts of the push buttons); an emergency stop removes the power; and a lamp H1 lights up when limit switch S3 detects that the cylinder is fully retracted. (14)

[20]

QUESTION 5

5.1 List, in the correct order, FIVE steps of a systematic fault-finding procedure. (5)

5.2 A solenoid valve does not switch when its coil is energised. Give FOUR possible causes. (4)

5.3 Explain the half-split method of fault-finding. (4)

5.4 State FOUR checks you would carry out on a hydraulic hose. (4)

5.5 Explain why a machine log book must be kept. (3)

[20]

TOTAL: 100

END OF QUESTION PAPER